

Assignment 4: Neural Network Verification with alpha,beta-CROWN

Reliable and Trustworthy Artificial Intelligence | June 2026

1. Model, Dataset, and Verification Property

Model. A small fully-connected network (MLP) for MNIST digit classification with the architecture 784 -> 64 -> 32 -> 10 (ReLU activations, no softmax). The network was trained for 5 epochs with the Adam optimiser (lr=1e-3) and achieves 96.31% accuracy on the MNIST test set. The model was exported to ONNX (opset 14) with an explicit Flatten layer in the graph so that alpha,beta-CROWN's built-in MNIST data loader, which returns 28x28 images, works directly without any preprocessing wrapper.

Dataset. The MNIST test set. Verification was performed on the first 100 samples (indices 0-99) with no normalisation (mean=0, std=1).

Verification property. L-infinity robustness: for each test image x with true label y , verify that $\arg \max f(x') = y$ for all x' satisfying $\|x' - x\|_\infty \leq \epsilon$, clipped to $[0,1]$. The perturbation radius $\epsilon = 0.01$ was chosen to match Assignment 3 (Marabou) exactly, enabling a direct comparison between the two tools.

2. Verification Results and Interpretation

alpha,beta-CROWN was run with the kfsb branching strategy, beta-CROWN with 20 iterations, and a 60-second per-sample timeout on a CPU (no GPU). Table 1 summarises the outcomes for all 100 samples.

Table 1. alpha,beta-CROWN results (100 MNIST samples, epsilon=0.01, L-inf)

Outcome	Count	%	Avg time (s)
safe-incomplete (CROWN bound sufficient)	97	97%	0.281
safe (BaB complete proof)	1	1%	0.567
unsafe-pgd (adversarial example found)	2	2%	0.008
timeout	0	0%	-
Total	100	100%	0.279

97 of 100 samples were verified safe by the CROWN linear-relaxation bound alone (safe-incomplete), meaning the bound-propagation step was already tight enough to prove robustness without any branch-and-bound splitting. One sample required a full BaB proof (safe). Two samples (indices 8 and 38) had adversarial examples found within the epsilon-ball by the PGD attack stage in under 12 ms each. These are images whose decision boundary lies close enough to the test point that a small perturbation can flip the prediction.

The 98% verified accuracy is remarkably high for $\epsilon=0.01$ on a plain MLP without adversarial training. This suggests the model has learned features that are inherently stable under small pixel perturbations at this scale.

3. Comparison: alpha,beta-CROWN vs Marabou

The same MLP architecture (784->64->32->10, ReLU) was used in Assignment 3 with Marabou. Table 2 compares both tools on the same task (eps=0.01, L-inf, 100 samples).

Table 2. alpha,beta-CROWN vs Marabou (same architecture, eps=0.01, 100 samples)

Metric	alpha,beta-CROWN	Marabou (A3)
Verified (safe)	98 / 100 (98%)	72 / 100 (72%)
Falsified (unsafe)	2 / 100 (2%)	28 / 100 (28%)
Timeout	0	0
Avg time / sample (s)	0.279	8.3
Speedup vs Marabou	~30x	1x
Model input format	ONNX (opset 14)	ONNX (opset 14)
Verification method	Bound prop + BaB	LP / SMT (ReLU split)

Speed. alpha,beta-CROWN is approximately 30x faster per sample. Most samples are resolved by CROWN in a single pass (~0.28 s), whereas Marabou's LP/SMT encoding requires iterative constraint solving for every ReLU neuron split.

Verified accuracy difference. alpha,beta-CROWN verifies 98% vs Marabou's 72%. Two factors contribute: (1) the models come from independent training runs (same architecture, different random seeds), so their decision boundaries differ; (2) alpha,beta-CROWN's integrated PGD attacker quickly discovers adversarial examples before invoking BaB, falsifying only 2 samples in milliseconds, while Marabou's exhaustive SMT search found adversarial examples for 28 samples. The discrepancy likely also reflects that the A3 model may have been trained on flattened inputs (no Flatten layer) while this model includes Flatten, producing slightly different learned representations.

Ease of use. alpha,beta-CROWN requires only a YAML config covering the model path, dataset, epsilon, and solver parameters. Marabou requires per-sample Python query construction with explicit bound variables. alpha,beta-CROWN also natively integrates a PGD attacker and prints a summary table automatically.

4. Strengths and Limitations of alpha,beta-CROWN

Strengths

+ Speed via bound propagation:

CROWN linear relaxation often proves robustness in a single forward/backward pass (safe-incomplete), making it orders of magnitude faster than SMT-based tools for small perturbation radii on well-trained models.

alpha,beta-CROWN won multiple VNN-COMP iterations on networks far larger than this MLP. It handles CNNs and residual nets that would be intractable for Marabou.

The built-in PGD attacker runs before BaB, finding counterexamples in milliseconds and avoiding expensive formal-proof attempts for clearly non-robust samples.

A single config file controls the full pipeline: model path, dataset normalisation, epsilon, branching strategy, and timeout.

Limitations

- Incomplete certificate (safe-incomplete):

When CROWN alone suffices, the result is sound but labelled incomplete. For a fully rigorous proof, BaB must terminate, which may not happen within the timeout for large epsilon or deep networks.

The tool is designed for GPU-accelerated tensor operations. On CPU it is still ~30x faster than Marabou for this model, but larger networks would benefit greatly from GPU availability.

Requires a specific conda environment with PyTorch, auto_LiRPA, onnx2pytorch, and several other packages. On Windows with non-ASCII paths, additional encoding workarounds were needed.